

Intelligent Driver Assistance with Accident – Prone Zone and Weather Alerts

A SYNOPSIS REPORT

Submitted by

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in partial fulfillment for the award of the degree of

B.Tech

IN

Artificial Intelligence and Data Science



Sant Dnyaneshwar Shikshan Sanstha's

**ANNASHEB DANGE COLLEGE OF ENGINEERING AND
TECHNOLOGY, ASHTA**

An Empowered Autonomous Institute

September ,2025

**ANNASAHEB DANGE COLLEGE OF ENGINEERING AND
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DECLARATION

I hereby declare that the Synopsis Report entitled, **“Intelligent Driver Assistance with Accident-Prone Zone and Weather Alerts”** submitted to Annasaheb Dange College of Engineering and Technology, Ashta, Maharashtra, INDIA in the partial fulfillment of the award of the Degree of **“B.tech”** in **“AI&DS”** is a bonafide work carried out by me. The material contained in this Project has not been submitted to any University or Institution for the award of any degree.

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1. Introduction:-

Road safety is one of the most pressing global challenges today. According to the World Health Organization (WHO), nearly 1.3 million people die each year due to road traffic accidents, and millions more suffer injuries or long-term disabilities. A major contributing factor to these accidents is the lack of timely awareness about accident-prone areas (blackspots), poor weather conditions, and unexpected road hazards such as potholes or sharp curves. With the rising number of vehicles and rapid urbanization, traditional road signs and static warnings are not sufficient to address real-time driving risks. The Accidental Intelligence Navigator (AIN) has been designed as an iOS-based mobile application that enhances road safety by providing real-time, proactive alerts to drivers. Unlike conventional navigation apps that only focus on shortest or fastest routes, AIN integrates multiple data streams such as GIS accident blackspot data, live weather updates, GPS tracking, and crowdsourced hazard reports to dynamically assess the risk on a driver's route. By doing so, the system not only navigates but also acts as an intelligent co-driver that helps prevent accidents before they occur. AIN leverages the full capabilities of modern smartphones—Core Location (GPS), MapKit, Core Motion sensors, and Firebase backend services—to deliver timely safety notifications through visual banners, voice prompts, and haptic feedback. For example, if a user is approaching a curve where multiple accidents have historically occurred, or if sudden rainfall is detected on their route, the system immediately issues an alert with enough lead time for corrective driving. This ensures that users can take safer decisions such as slowing down, rerouting, or preparing for hazards in advance. Furthermore, AIN incorporates crowdsourcing features, allowing drivers to report hazards like accidents, potholes, or roadblocks. These reports are validated, aggregated, and shared with other drivers in real-time, making the system self-improving and community-driven. Privacy is also a key focus; sensitive data is minimized, with most processing handled on-device, ensuring user trust and compliance with privacy standards. By blending machine learning models, GIS analysis, and smartphone sensor fusion, AIN represents a significant step towards next-generation road safety systems. Its potential lies in reducing the number of accidents, minimizing traffic delays, and ultimately saving lives. The system can be scaled globally, but the MVP rollout will focus on India, where road accident rates are among the highest in the world, thereby addressing a highly relevant and impactful problem.

2. Objective & Scope of Project:-

- **Objectives:-**

- a. Detect accident-prone and recently reported accident locations using real-time GPS and historical incident data.
- b. Alert users via push notifications, voice, and haptics when approaching hazardous zones.
- c. Provide early warnings for weather-related risks (e.g., rain) and road geometry risks (e.g., sharp turns, steep grades).
- d. Reduce accident risk and traffic delays through timely, contextual alerts and safer rerouting.
- e. Deliver a user-friendly, privacy-preserving iOS app for real-time safety notifications.

- **Scope:-**

- a. Flutter app (Android and iOS) with localization ready for India-first rollout.
- b. City and highway navigation; live risk scoring per road segment.
- c. Cloud backend (Firebase) for blackspot data, user reports, and model updates.

3. Literature Survey:-

1. A Smart GPS-Based IoT System for Blackspot Notification:-

The paper proposes an IoT-based system that uses GPS and GSM modules to notify drivers of accident blackspots through audio alerts. The system integrates Arduino microcontrollers with GPS coordinates of known blackspots to provide real-time driver notifications. Tested on notorious blackspots in Kenya, including Salgaa and Mombasa Road, with successful detection and alerts.

2. Investigation of Road Accident Zone and Suggestion for Preventive Measures by Using GPS and GIS Technology in Thanjavur, Tamil Nadu, India:-

The study investigates road accident zones in Thanjavur using GPS and GIS technology for hotspot detection and preventive measures. Accident data analyzed across parameters: police station-wise, vehicle type, time, road type, and location. Hotspots identified include Soorakottai bus stop, Echankottai, Kulichapatu cut road, and others, based on speeding and poor infrastructure.

3. AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM :-

The paper proposes a mobile application integrated with Gaussian Mixture Modelling (GMM) and Support Vector Machine (SVM) classifiers to predict and alert users of road accident blackspots in the Vijayawada region. Accident data from the Road Transportation Authority (RTA) was preprocessed, features selected, and oversampled using SMOTE to balance severity classes. The model classifies accidents into three categories: fatal, serious injury, and general injury, with SVM achieving the highest accuracy (96.66%).

4. Prediction and Identification of Black Spots to Reduce the Severity of Accidents Using Machine Learning Algorithms:-

The study applies machine learning models (Decision Tree, Random Forest, Gradient Boosting, (XGBoost) to predict road accident severity and identify blackspots in Tamil Nadu, India. Dataset includes five years of accident records (2017–2021) with factors such as road type, vehicle category, time, and weather. XGBoost achieved the highest accuracy (98.3%) in predicting accident severity and blackspot locations.

5. Identification and Mitigation of Road Accident Black Spots Using Data Analytics and Machine Learning:-

The study integrates data analytics and ML models (Logistic Regression, Random Forest, Gradient Boosting) to identify accident-prone blackspots and suggest mitigation measures. Dataset includes three years of accident records from highways in Tamil Nadu, India, covering factors such as time, weather, and driver behavior. Random Forest achieved strong predictive performance in identifying high-risk zones compared to other models.

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Sr. No	Title	Author	Key Points
1	A Smart GPS-Based IoT System for Blackspot Notification	1. Billy Owire, 2. Bernard Shibwabo Kasamani	- The paper proposes an IoT-based system that uses GPS and GSM modules to notify drivers of accident blackspots through audio alerts. - The system integrates Arduino microcontrollers with GPS coordinates of known blackspots to provide real-time driver notifications. - Tested on notorious blackspots in Kenya, including Salgaa and Mombasa Road, with successful detection and alerts.
2	Investigation of Road Accident Zone and Suggestion for Preventive Measures by Using GPS and GIS Technology in Thanjavur, Tamil Nadu, India	1. M. Kannan, 2. G. Aakash	- The study investigates road accident zones in Thanjavur using GPS and GIS technology for hotspot detection and preventive measures. - Accident data analyzed across parameters: police station-wise, vehicle type, time, road type, and location. - Hotspots identified include Soorakottai bus stop, Echankottai, Kulichapatu cut road, and others, based on speeding and poor infrastructure.
3	AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM	1. M. Sobhana, 2. V. Krishna Rohith, 3. T. Avinash, 4. N. Malathi, 5. Smitha Chowdary	- The paper proposes a mobile application integrated with Gaussian Mixture Modelling (GMM) and Support Vector Machine (SVM) classifiers to predict and alert users of road accident blackspots in the Vijayawada region. - Accident data from the Road Transportation Authority (RTA) was preprocessed, features selected, and oversampled using SMOTE to balance severity classes. - The model classifies accidents into three categories: fatal, serious injury, and general injury, with SVM achieving the highest accuracy (96.66%).

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4	Prediction and Identification of Black Spots to Reduce the Severity of Accidents Using Machine Learning Algorithms	1. Arun Kannan, 2. M. Sivaprakash, 3. P. Sudhakar, 4. S. Saravanan	• The study applies machine learning models (Decision Tree, Random Forest, Gradient Boosting, XGBoost) to predict road accident severity and identify blackspots in Tamil Nadu, India. • Dataset includes five years of accident records (2017–2021) with factors such as road type, vehicle category, time, and weather. • XGBoost achieved the highest accuracy (98.3%) in predicting accident severity and blackspot locations.
5	Identification and Mitigation of Road Accident Black Spots Using Data Analytics and Machine Learning	1. R. Bhuvaneshwari, 2. K. Sathesh Kumar, 3. R. Karthik, 4. P. Chandrasekaran	• The study integrates data analytics and ML models (Logistic Regression, Random Forest, Gradient Boosting) to identify accident-prone blackspots and suggest mitigation measures. • Dataset includes three years of accident records from highways in Tamil Nadu, India, covering factors such as time, weather, and driver behavior. • Random Forest achieved strong predictive performance in identifying high-risk zones compared to other models.

4. Design and Project Model:-

- **System Context**

- a. **Client App:**

Built using Dart and Flutter with plugins like geolocator, google_maps_flutter, background_fetch, and flutter_local_notifications.

- b. **Backend:**

Firebase Authentication, Firestore (hazard and blackspot data), Firebase Cloud Functions (risk aggregation), Firebase Cloud Messaging (push alerts).

- c. **Data Sources:**

Historical blackspots (GeoJSON), crowdsourced hazard reports, live weather API (Open-Meteo/Apple WeatherKit), optional traffic density.

- **High-Level Architecture**

- **Location & Telemetry Layer:** GPS streams (lat, lon, speed, heading) via Geolocator plugin.
- **Map & Routing:** Google Maps integration; overlays for risk zones; polyline segmentation ($\approx 50\text{--}200$ m).
- **Risk Engine (client + cloud):**
 - Features: Distance to nearest blackspot, segment crash density, speed vs. limit, curvature, grade, weather (rain), recent crowd reports, driver behavior flags.
 - Model: On-device TensorFlow Lite or Firebase ML for low-latency risk (0–10 score). Periodic cloud aggregation for model refresh.
- **Alerting & UX:** Haptics + voice prompts when risk $>$ threshold and ETA $<$ configurable horizon (e.g., 20–30 s). One-tap “Why?” sheet explaining factors; “Report hazard” action.
- **Privacy & Security:** All PII minimized; location processed on-device when possible; end-to-end TLS; Firebase Security Rules; user control for data sharing.

- **Data Model (Core entities):-**

- a. Blackspot: id, geometry (polyline/point), severity, source, last_updated.
- b. Hazard Report: : id, type, location, confidence, reporter_hash, expires_at.
- c. Segment Risk: segment_id, features[], risk_score, updated_at.
- d. User Event: speed, braking_flag, over_speed_ratio, timestamp.

- **Key Screens:-**

- a. **Home/Go:** destination search; Start Navigation.
- b. **Map + Risk Overlay:** route polyline, color-coded risk (green→red).
- c. **Alerts:** banner + voice; details bottom-sheet.
- d. **Report Hazard:** quick forms with type (pothole, crash, sharp turn), photo optional.
- e. **History:** trips, alerts received, opt-in analytics.

5. Methodology:-

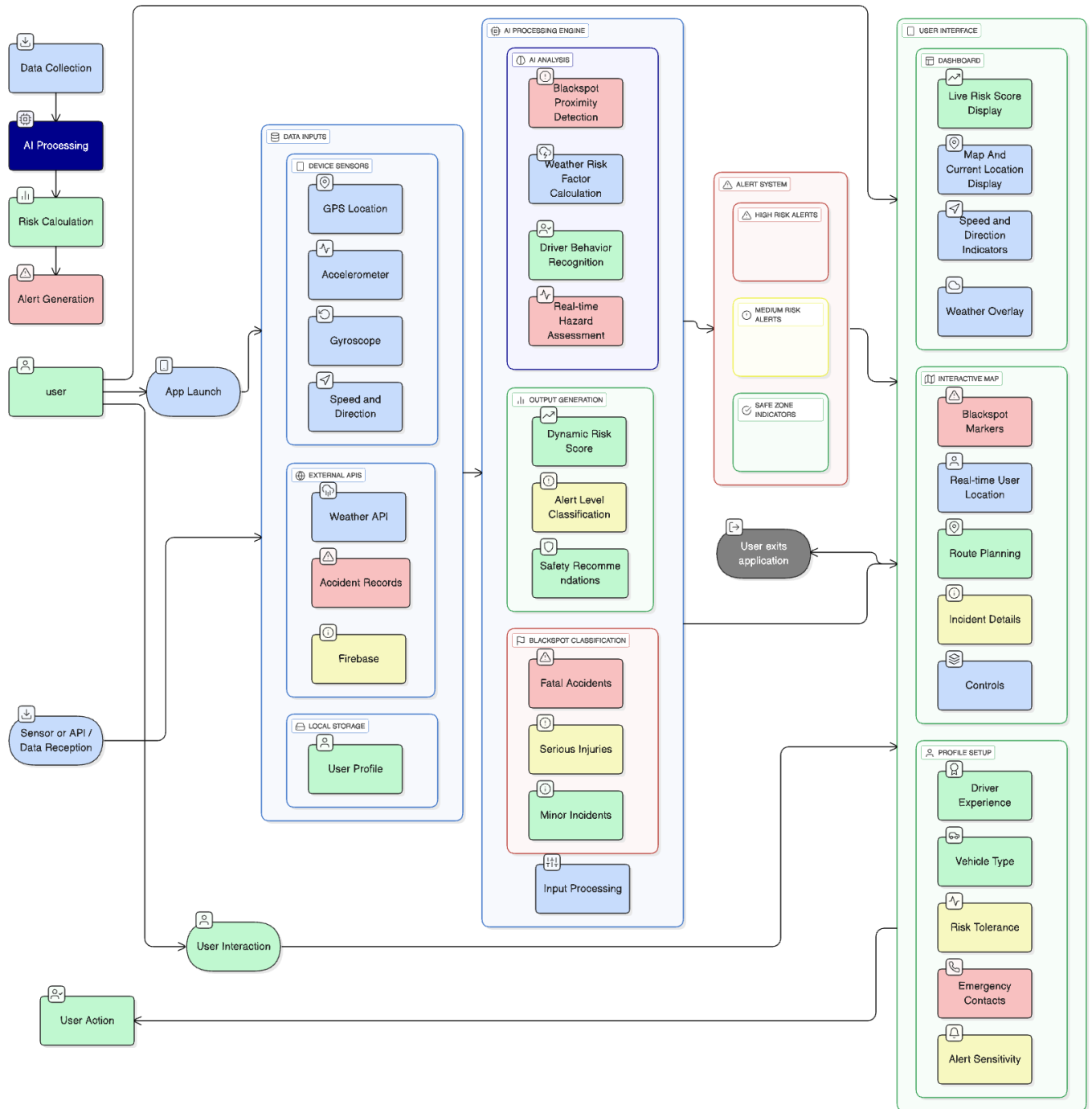


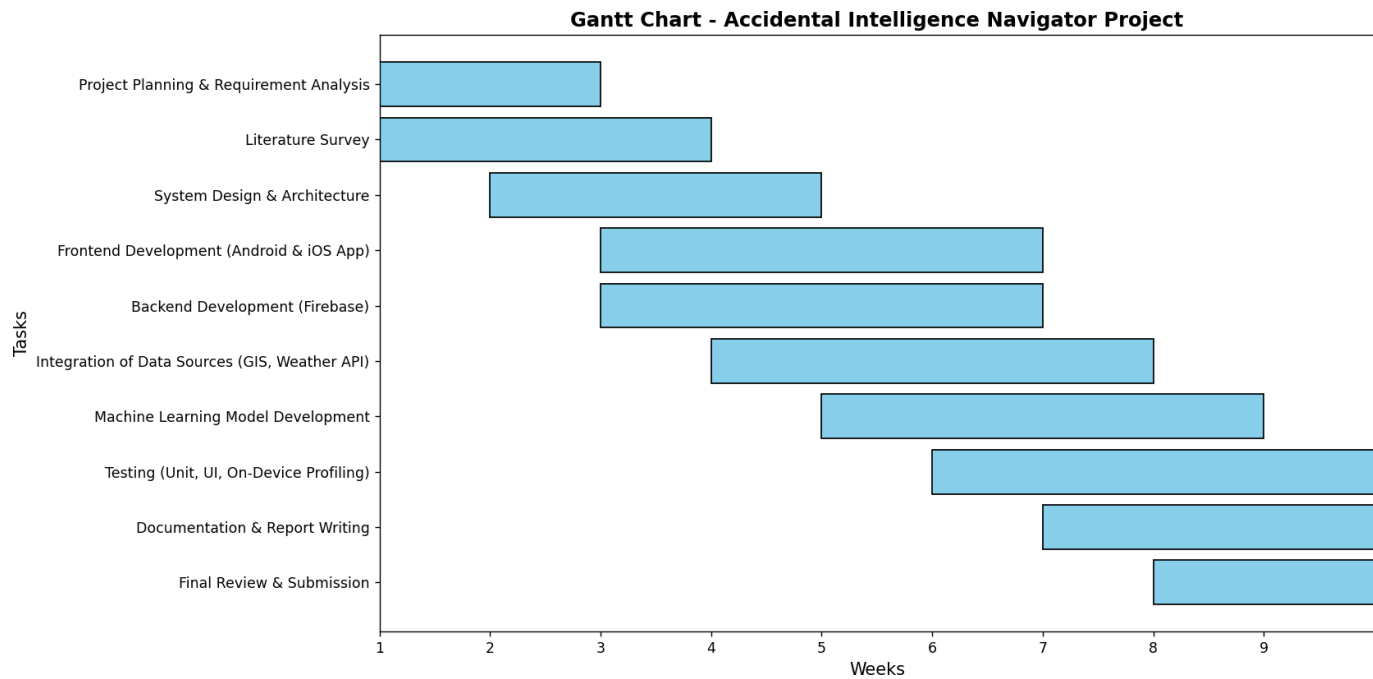
Fig. Workflow

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- a) Data Ingestion:** Import GIS blackspots; sync to Firestore; pull live weather per route.
- b) User Trip Setup:** Profile (age optional), vehicle type, allowed alerts.
- c) Routing & Segmentation:** Compute route; split into segments; attach features.
- d) Risk Scoring:** On-device model outputs risk 0–10 per segment; cloud updates longer-term risk maps.
- e) Real-Time Alerts:** If $\text{distance_to_hazard} < \text{threshold}$ and $\text{risk} > \text{policy}$, trigger voice/haptic/banner.
- f) Behavior Monitoring:** Detect aggressive events (hard brake, speeding) to adapt risk.
- g) Crowd Reporting:** Users can add/update hazards; server validates and decays stale items.
- h) Evaluation:** A/B of alert policies; measure false positives/negatives; iterate thresholds.

6. Gantt Chart :-

- Timeline of the Project:-



7. Hardware and Software Used:-

- **Hardware:-** Smartphone (Android/iOS) with GPS, motion sensors; optional car mount; reliable internet.
- **Software:-**
 - a. **Client:** Flutter 3+, Dart, geolocator, google_maps_flutter, background_fetch, flutter_local_notifications.
 - b. **Backend:** Firebase Auth, Firestore, Cloud Functions (Node.js), Cloud Messaging, Firebase Crashlytics & Analytics.
 - c. **DevOps:** GitHub, Fastlane, Firebase Test Lab, Crashlytics dashboards.

8. Testing Technologies Used:-

- a) **Unit Tests:** Flutter test package for feature extractors, risk inference, data mappers.
- b) **Widget Tests:** Flutter widget testing for navigation flows and alert banners.
- c) **On-Device Profiling:** Flutter DevTools for performance, memory, and battery profiling.
- d) **A/B & Analytics:** Firebase Remote Config + Analytics for threshold tuning.
- e) **Beta & Crash:** Firebase App Distribution; Crashlytics for stability.
- f) **Field Trials:** Drive tests covering rain/clear, highway/city, day/night; compare alert logs with ground truth (known blackspots, simulated hazards).

9. Contribution of Project:-

- a)** Early warnings for blackspots and weather-risk segments during daily commutes.
- b)** Privacy-respecting crowdsourcing of hazards to keep maps fresh.
- c)** By alerting drivers about high-risk areas, the app indirectly helps emergency services by reducing accident loads.
- d)** Directly to improving safety in a country with one of the highest global road accident rates.

10.Role of Members Individually :-

Sr. No	Frontend Development	Backend Development	Research	Document
1.	Prem Kadam	Saurabh Mali	Mugdha Patil	Sanika Mohite
2.	Mugdha Patil	Sanika Mohite	Saurabh Mali	Prem Kadam

11.Base Papers from 2020 and Above:-

1. AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM:-

The paper “*AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM*” by Sobhana, Krishna Rohith, Avinash, Malathi, and Smitha Chowdary focuses on reducing road accidents by developing a mobile application that leverages artificial intelligence. The study highlights that Andhra Pradesh, particularly the Vijayawada region, has recorded a high number of road accidents over the years, making blackspot detection and alerting systems crucial. Accident data collected from the Road Transportation Authority (RTA) was analyzed and categorized into three severity levels: fatal, severe injury, and general injury. Before training the models, the dataset underwent preprocessing steps such as handling missing values, feature selection, and applying SMOTE to resolve class imbalance. The authors used a hybrid model where Gaussian Mixture Modelling (GMM) identified accident-prone clusters, and Support Vector Machine (SVM) classified accident severity. Among the tested classifiers, SVM achieved the best performance with an impressive accuracy of 96.66%, demonstrating its effectiveness in predicting accident severity.

The novelty of the work lies in the integration of this predictive model into a cloud-connected Android application. The app records user information such as age, gender, and vehicle type, and tracks real-time location using GPS. By mapping routes, the system can alert users if their journey passes through a blackspot, and it issues live safety alerts as they approach high-risk areas. This makes the solution practical for daily use by commuters. The deployment on Google Cloud ensures scalability, enabling seamless data access and updates.

Despite its promising results, the paper also acknowledges certain limitations. The reliance on historical accident records may not capture real-time factors like weather, road conditions, or driver behavior. The use of SMOTE, while balancing data, may risk overfitting and reduce generalizability to unseen cases. Moreover, accident blackspots change dynamically with road repairs or new infrastructure, so the model requires periodic updates. The system also raises privacy concerns, as it stores sensitive user data and tracks their movements.

Overall, this research demonstrates how combining AI techniques with mobile technology can create an effective accident prevention system. It not only predicts the likelihood of accidents at specific locations but also provides real-time support to users, potentially reducing fatalities and injuries. This paper stands out as a practical example of how machine learning models like GMM and SVM can move beyond theoretical research into real-world road safety applications.

12. References IEEE:-

1. “A Smart GPS-Based IoT System for Blackspot Notification” , “Billy Owire, Bernard Shibwabo Kasamani” , IEEE ,(2021)
2. “Investigation of Road Accident Zone and Suggestion for Preventive Measures by Using GPS and GIS Technology in Thanjavur, Tamil Nadu, India” , “M. Kannan, G. Aakash” , ResearchGate(2024).
3. “AI Enabled Accident Black Spot Alerting Mobile System to Enhance Road Safety Using GMM-SVM” , “M. Sobhana, V. Krishna Rohith” ,IEEE(2023).
4. “Prediction and Identification of Black Spots to Reduce the Severity of Accidents Using Machine Learning Algorithms” , “Arun Kannan, M. Sivaprakash” ,IEEE(2024).
5. “Identification and Mitigation of Road Accident Black Spots Using Data Analytics and Machine Learning” , “R. Bhuvaneshwari, Sathesh Kumar” ,ScienceDirect (2023).